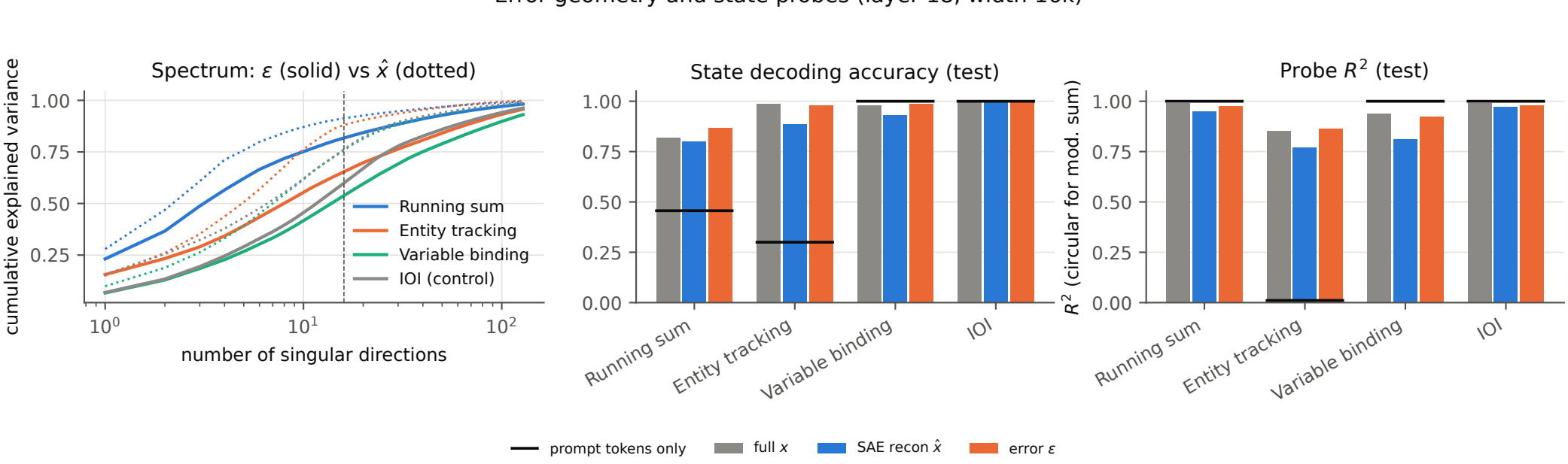
Error geometry and state probes (layer 18, width 16k)
Spectrum: ε (solid) vs x̂ (dotted)
cumulative explained variance
1.00
0.75
0.50
0.25
10^0
10^1
10^2
number of singular directions
Running sum
Entity tracking
Variable binding
IOI (control)
State decoding accuracy (test)
1.00
0.75
0.50
0.25
0.00
Running sum
Entity tracking
Variable binding
IOI
Probe R^2 (test)
R^2 (circular for mod. sum)
1.00
0.75
0.50
0.25
0.00
Running sum
Entity tracking
Variable binding
IOI
prompt tokens only
full x
SAE recon x̂
error ε